

Multi-time scale optimization scheduling of microgrid considering source and load uncertainty

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ARTICLE INFO

Keywords:

Multi-time scale
Microgrid
Optimal scheduling
Distributionally robust optimization
Distributed model predictive control

ABSTRACT

With the increase of renewable energy penetration in microgrid and the stochasticity of customer load, microgrid faces new difficulties in maintaining the smooth power of contact lines and system economy when achieving optimal scheduling. In order to balance the accuracy, economy and robustness of microgrid scheduling better, a multi-time scale optimal scheduling strategy for microgrids considering the uncertainty of source and load is proposed. In the day-ahead scheduling stage, a two-stage distributionally robust optimal scheduling model is established with the objective of minimizing the comprehensive day-ahead scheduling cost of the microgrid, and the optimal day-ahead scheduling solution is found under the probability distribution of the worst scenario. In the intra-day scheduling stage, based on the prediction value of source and load with higher accuracy, a distributed model predictive control method is used to build an intra-day rolling optimization and real-time adjustment optimal scheduling model, which ensures the effective implementation of the day-ahead plan and fully suppresses power fluctuation of the contact line. The simulation results show that the proposed multi-time scale optimal scheduling method can not only maintain the smooth power of contact lines, but also achieve robust and economic operation of the microgrid.

1. Introduction

As the electrification rate of energy systems increase, the importance of renewable energy sources is coming to the fore. The decline in fossil fuel consumption will be accompanied by a rapid increase in renewable energy. The installed capacity of wind and solar energy is expected to increase more than 15 times by 2050 compared to 2019 [1,2]. With the goal of "carbon peaking" and "carbon neutrality", it has gradually become a consensus among the government and the public to vigorously develop distributed power sources and reduce the excessive use of fossil fuels [3]. As an advanced stage of distributed power supply and an important part of the future energy internet, microgrid provides an effective way to improve the utilization of clean energy [4,5]. Microgrid, which contains renewable energy, various energy transmission devices and energy storage devices, can achieve efficient and graded utilization of energy by planning and scheduling the output of each unit to meet the demand of user-side load [6–8].

As an important part of microgrid energy management, optimal scheduling of microgrid can guarantee the economic and safe operation of microgrid on the basis of satisfying the operational constraints of

equipment within the system [9,10]. However, the volatility of renewable energy sources and the diversity of users' energy usage inevitably exist, which make the microgrid source-load sides have strong uncertainty, so uncertain optimization methods are applied to the microgrid to reduce the impact of uncertainty of source and load [11,12].

Stochastic optimization generates scenarios by using prior empirical probability distributions, and solve for the scheduling solution with the lowest cost expectation [13]. In [14], scenario-based stochastic approach is used to manage uncertainties arising from renewable energy, electricity, market price, demands in multi-carrier microgrids. The authors of [15,16] present a stochastic model to deal with the uncertainties of loads and renewable energy. However, stochastic optimization needs to generate a large number of scenarios based on the probability distributions, which may cause severe computational burdens [13–16]. Compared to the stochastic programming method, robust optimization uses various uncertainty sets to characterize uncertain variables in order to find the optimal solution for the worst-case scenario. Uncertainty set can effectively reflect the uncertainties of source and load. Literatures [17,18] use a two-stage robust optimization model to derive the optimal dispatch plan for microgrids to deal with the

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<https://doi.org/10.1016/j.epsr.2022.109037>

Received 4 October 2022; Received in revised form 9 November 2022; Accepted 26 November 2022

Available online 7 December 2022

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stochastic nature of distributed power output in the microgrid system. In [19], a double-layer robust dispatch model with an improved uncertainty set which applies a neural network is proposed, so as to reduce the impact of uncertainty in electricity demand and electric vehicles in microgrids. Although robust optimization does not need to generate a large number of scenarios, the results are often too conservative. Distributionally robust optimization method solves the optimal solution under the worst probability distribution of uncertain variables, which can effectively achieve the balance of economy and robustness [20]. Considering the uncertainty of distributed power supply, literature [21] proposes a two-stage distributionally robust method to operate energy hub with an energy storage system, which can help energy hub system consume energy more economically. The authors of [22] propose a distributionally robust dispatching model for microgrid considering the time-delay of pipelines and the uncertainty of renewable energy.

Most of the above studies have analyzed the uncertainties of renewable energy and load. However, many current studies only develop offline full time scheduling plans based on the day-ahead 24 h long time scale forecast information at one time, which may make the dispatching results differ significantly from the actual operating state of the microgrid. What's more, day-ahead scheduling alone may not fully reflect the impact of unplanned instantaneous power fluctuations of source and load on the energy management of the microgrid, which is detrimental to the stability of the microgrid's interaction with external networks such as the grid and the gas network [23].

The accuracy of the prediction value of source and load has the characteristic of improving with the decrease of the time scale, so multi-time scale optimization is applied to the field of microgrid dispatching to maintain the smooth power of contact lines between microgrid and external networks [24–26]. In [27], a multi-time scale scheduling strategy was proposed for microgrid, in which the system is able to pre-allocate the capacity of the system before the day and adjust the day-ahead scheduling plan according to the real-time capacity of renewable energy sources during the day. Because the Model Predictive Control (MPC) has the characteristics of rolling optimization and feedback correction, the method can be applied in multi-time scale optimization to meet the requirements of real-time adjustment and reduce the fluctuation of renewable energy and load [28]. The authors of [29,30] establish optimization problems with energy balance constraint based on MPC to minimize the impact of load fluctuations. In [31], an optimal dispatch model is constructed by using MPC based on real-time energy price and load forecast data in order to minimize the system operation cost. However, for the microgrid connected with a large number of distributed generation units and adjustable components, centralized MPC scheduling has the shortcoming of large calculation volume, low reliability and being hard to expand. The proposal of Distributed Model Predictive Control (DMPC) can effectively solve the shortcomings of MPC. Literature [32] coordinates the power output of each part of the microgrid based on DMPC, which can better respond to the load fluctuations and improve the system robustness. In [33], a DMPC strategy is used to deal with the uncertainties in both supply and demand sides in microgrid, the results show that the proposed strategy can not only improve global distributed generation utilization efficiency, but also guarantee the overall benefit. The authors of [34,35] use DMPC to control the load frequency and energy scheduling of the microgrid respectively, improving the control performance through the mutual cooperation of subsystems.

Although the above-mentioned literature uses source-load prediction values of different time scales to reduce the influence of uncertainties, they all construct models based on the day-ahead deterministic optimal scheduling results, which makes the overall scheduling strategy overly dependent on the accuracy of source load prediction. In actual operation, the day-ahead scheduling plan of the microgrid needs to maintain certain robustness, so that the power fluctuations required by the system for adjustment in the intra-day phase are minimized, avoiding excessive power adjustments in the intra-day plan due to the presence of extreme

scenarios.

To address the above problems, this paper proposes a multi-time scale optimal scheduling strategy for microgrid: in the day-ahead scheduling stage, considering the uncertainty of scenery load, a two-stage distributionally robust day-ahead optimal scheduling model is constructed with the objective of minimizing the comprehensive daily operation cost of the microgrid to avoid excessive intra-day costs due to adjustment; in the intra-day scheduling stage, in order to realize the tracking of the day-ahead scheduling plan of the microgrid and improve the shortcomings of the traditional MPC solution with high order and large dimension, an intra-day rolling optimization and real-time adjustment model is constructed based on the DMPC method, which enables real-time adjustments effectively to track the day-ahead scheduling plan, reduce the power fluctuations caused by source load uncertainty, thus ensure the stability of the contact line power, and finally realize the robustness and economy of the microgrid operation.

The rest of the paper is structured below. The physical model of the microgrid and problem description is provided in Section 2, the multi-time scale optimal scheduling models for microgrids are given in Section 3, the results of case study are discussed in Section 4, and the final conclusion is made in Section 5.

2. Microgrid model structure and problem description

2.1. Microgrid model structure

This paper is based on the integrated energy structure system of "supply", "transfer", "storage" and "use", where renewable energy is introduced at the source, and the transmission devices and energy storage devices are used to meet the diversified load demands of electricity, heat and gas. The microgrid is shown in Fig. 1.

2.2. Problem description

The volatility of wind power and photovoltaic, as well as the randomness of energy use by users, make the prediction of both source and load inaccurate, thus affecting the stability of the contact line and increasing the scheduling cost. It is also necessary to consider extreme scenarios to ensure the robustness of microgrid scheduling. Based on this, this paper improves the prediction accuracy by reducing the time scale, and maintains system stability and scheduling economy by real-time adjustment. The problem description is shown in Fig. 2.

The multi-time scale framework of the microgrid established in this paper is shown in Fig. 3: it mainly contains three levels: day-ahead two-stage distributionally robust optimization scheduling, intra-day rolling optimization scheduling and real-time adjustment. The implementation process of the whole multi-time scale scheduling strategy is as follows: the day-ahead scheduling has a scale value of 1 h and is formulated every 24 h; the intra-day rolling optimization scheduling stage has a scale value of 15 min and uses 4 h as the control time domain for rolling optimization scheduling, but only sends the first 15 min scheduling instruction of each rolling optimization to the next stage; the real-time adjustment optimal scheduling takes 5 min as the index value to execute further adjustment based on the intra-day rolling optimization plan.

3. Multi-time scale optimal scheduling model for microgrids

3.1. Day-ahead two-stage distributionally robust optimal scheduling

The combined electricity, heat and gas microgrid including wind and photovoltaic has multiple uncertainties, and it is difficult for deterministic optimization methods to effectively cope with 24 h-scale scheduling. At the same time, traditional uncertainty optimization methods have certain one-sidedness and limitations: stochastic optimization generates scenarios through probability distribution, which is too

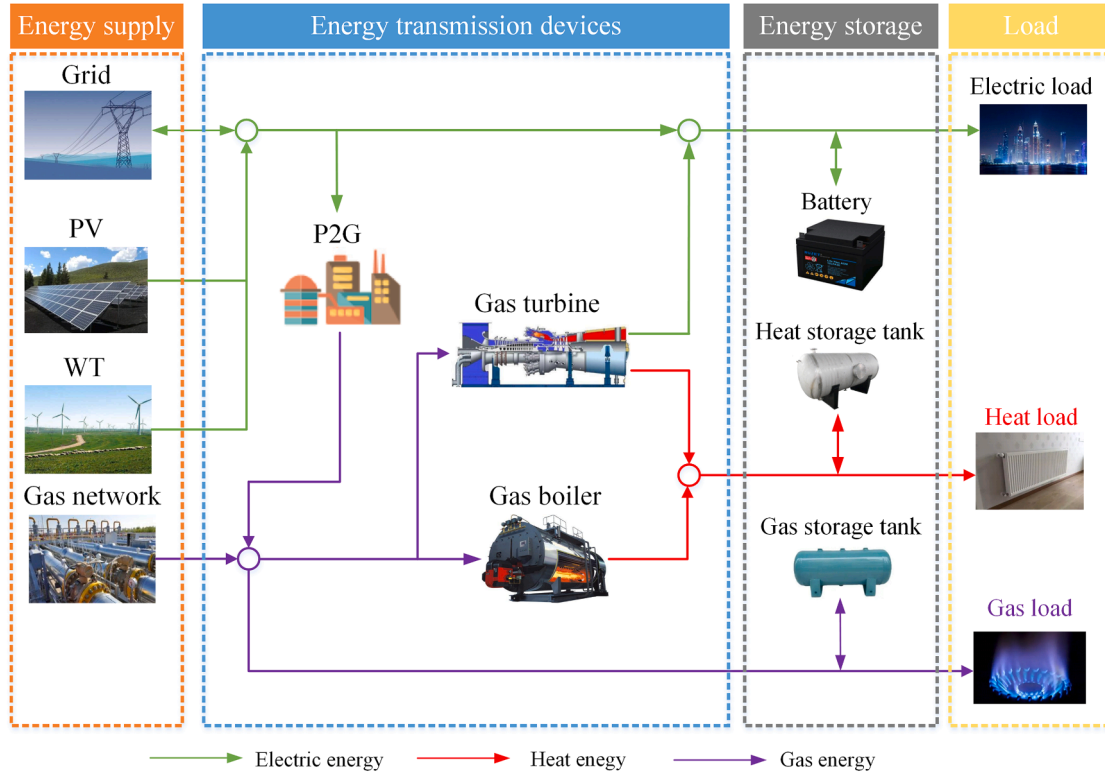


Fig. 1. Electricity, heat and gas combined microgrid structure.

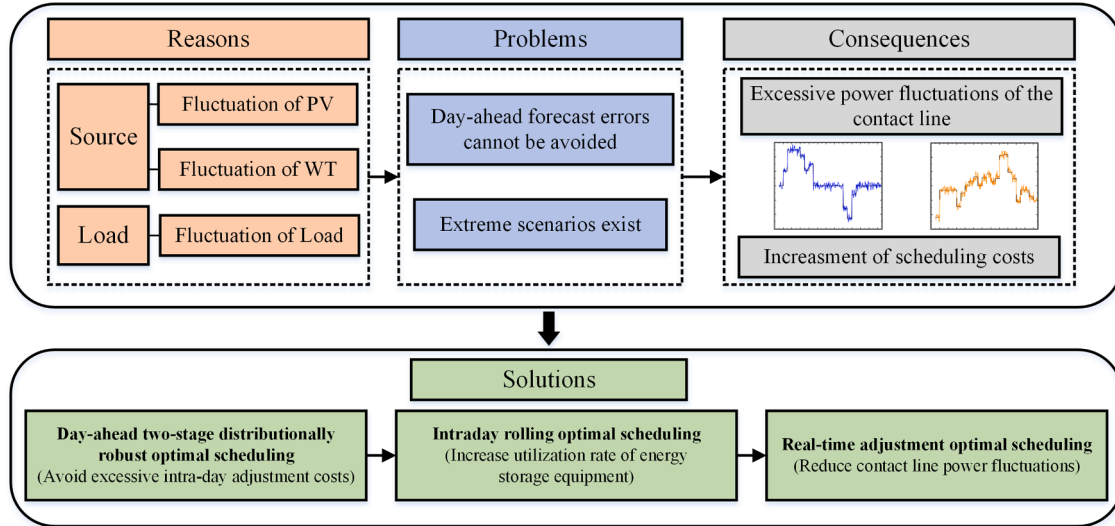


Fig. 2. Problem description.

computationally intensive and difficult to obtain accurate probability distribution functions in practice; robust optimization only considers the optimal solution of the model under the worst scenario, which makes the scheduling scheme too conservative and less economical.

Therefore, to avoid the one-sidedness of the day-ahead scheduling scheme, this paper constructs a day-ahead two-stage distributionally robust optimal scheduling model using a large amount of historical data existing in actual scheduling to characterize uncertain parameters and their probability distribution confidence sets. The first stage of the model develops the scheduling plan of each unit and equipment with the comprehensive operating cost of the microgrid as the target, and the second stage adjusts the scheduling plan of the first stage under the

probability distribution of the worst scenario of scenic load. The result ensures that the scheduling plan can meet the safe and stable operation requirements of the microgrid under all possible worst scenarios.

The objective function of the day-ahead two-stage distributionally robust optimal scheduling model is shown in Eq. (1) to develop the optimal scheduling plan under the probability distribution of the worst wind, solar and load scenario.

$$\min \left\{ [C_{buy} + C_{CO_2} + C_{om}] + \max_{\{p_n\} \in \Omega} \left\{ \sum_{n=1}^N p_n \min \left\{ \sum_{t=1}^T \sum_{s=1}^N \pi_t^s \Delta P_{t,n}^s \right\} \right\} \right\} \quad (1)$$

where C_{buy} is the energy purchase cost of the microgrid; C_{CO_2} is the

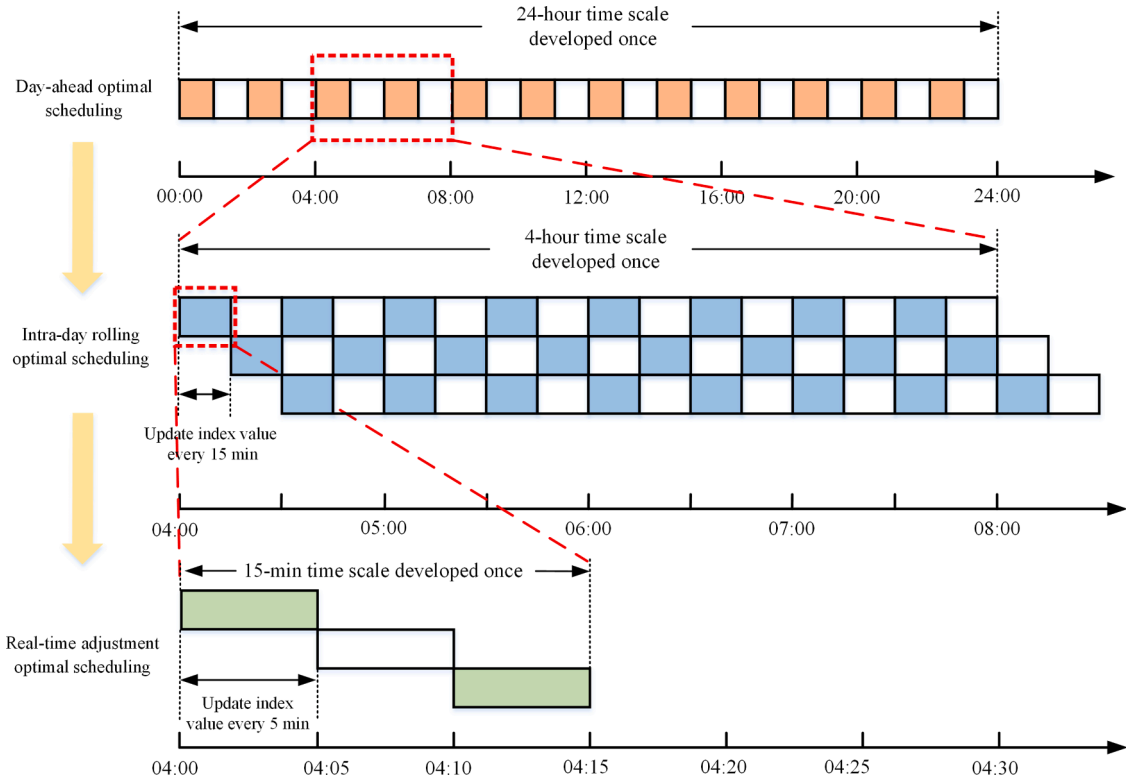


Fig. 3. The relationship between day-ahead, intra-day and real-time scheduling phases.

environmental emission control cost of the microgrid; C_{om} is the maintenance cost of gas turbine, gas boiler, P2G and energy storage devices in the microgrid; Ω is the probability distribution feasible domain of the scenario; p_n is the probability expectation value of the n th specific scenario; s represents gas turbine, gas boiler, and P2G in the microgrid; π_t^s is the adjustment cost of gas turbine, gas boiler, and P2G at time t ; $\Delta P_{t,n}^s$ is the n th scenario of the adjusted power magnitude of the s th device at moment t .

In Eq. (1), the internal problem(max-min) is mainly responsible for solving the worst probability distribution of scenic load scenarios when the total adjustment cost of each distributed energy device of the microgrid is minimized in the reschedule phase; the external problem (min-max-min) solves the microgrid scheduling scheme with the lowest total cost under the worst probability distribution of scenic load scenarios.

(1) Energy purchase costs

$$C_{buy} = \sum_{t=1}^T (\lambda_t^{pbuy} P_t^{buy} - \lambda_t^{psell} P_t^{sell}) \Delta t + \sum_{t=1}^T \lambda_t^{gbuy} G_t^{buy} \Delta t \quad (2)$$

where λ_t^{pbuy} , λ_t^{psell} and λ_t^{gbuy} are the prices of electricity purchased, electricity sold and gas purchased on the microgrid at time t respectively; P_t^{buy} , P_t^{sell} and G_t^{buy} are the amounts of electricity purchased, electricity sold and gas purchased on the microgrid at time t respectively.

(2) Environmental emission control costs

$$C_{CO_2} = \sum_{t=1}^T \varepsilon (P_t^{gt} + H_t^{gt}) \gamma_{gt} \Delta t + \sum_{t=1}^T \varepsilon H_t^{gb} \gamma_{gb} \Delta t + \sum_{t=1}^T \varepsilon P_t^{buy} \gamma_{grid} \Delta t \quad (3)$$

where P_t^{gt} and H_t^{gt} are the electricity and heat production of the gas turbine at time t respectively; H_t^{gb} is the heat production of the gas boiler at time t ; γ_{grid} , γ_{gt} and γ_{gb} are the CO₂ emission factors of the grid, gas turbine and gas boiler respectively; ε is the CO₂ treatment cost factor.

(3) Energy transmission equipment and energy storage equipment maintenance costs

$$C_{om} = \sum_{t=1}^T (\mu_{gt} P_t^{gt} + \mu_{gb} H_t^{gb} + \mu_{p2g} G_t^{p2g}) \Delta t + \sum_{t=1}^T [\nu_{es} (P_t^{ch} + P_t^{dis}) + \nu_{hs} (H_t^{ch} + H_t^{dis}) + \nu_{gs} (G_t^{ch} + G_t^{dis})] \Delta t \quad (4)$$

where, G_t^{p2g} are the gas production volume of the P2G equipment at time t ; μ_{gt} , μ_{gb} and μ_{p2g} are the maintenance costs of gas turbine, gas boiler, and P2G respectively; ν_{es} , ν_{hs} and ν_{gs} are the maintenance costs of battery, heat storage tank and gas storage tank respectively; P_t^{ch} and P_t^{dis} are charge and discharge power of the battery at time t ; H_t^{ch} and H_t^{dis} are charge and discharge power of the heat storage tank at time t ; G_t^{ch} and G_t^{dis} are the filling and releasing volumes of gas storage tanks at time t .

The constraints of the day-ahead scheduling model of the microgrid not only need to meet operation constraints of each device, the energy storage device charging and discharging constraints, the contact line delivery constraints [36–38], but also need to meet energy balance constraints as follows:

$$P_t^{buy} + P_t^{wt} + P_t^{pv} + P_t^{gt} + P_t^{dis} = P_t^{sell} + P_t^{p2g} + P_t^{ch} + P_t^L \quad (5)$$

$$H_t^{gb} + H_t^{gt} + H_t^{dis} = H_t^{ch} + H_t^L \quad (6)$$

$$G_t^{\text{buy}} + G_t^{\text{P2G}} + G_t^{\text{dis}} = G_t^{\text{gt}} + G_t^{\text{gb}} + G_t^{\text{ch}} + G_t^{\text{L}} \quad (7)$$

where, P_t^{L} is electrical load demand of the microgrid at time t ; H_t^{L} is the heat load demand of the microgrid at time t ; P_t^{P2G} are the electricity consumption of the P2G equipment at time t , respectively; G_t^{gt} and G_t^{gb} are the volumes of natural gas consumed by gas turbines and gas boilers at time t ; G_t^{L} is the gas load demand of the microgrid at time t .

to make the intra-day scheduling follow the day-ahead plan better, the start-stop penalty cost of each energy storage equipment is minimized as the objective function. Moreover, the contact line power fluctuation and storage energy adjustment amount are added to the objective function as penalty functions. The objective function for intra-day rolling optimal scheduling is shown in Eq. (11).

$$\min \sum_{t=T}^{T+k\Delta T} \left\{ \sum_{n=1}^N [I_n(t) - I_{n,\text{dh}}(t)] \cdot C_n + \sigma (P_{\text{buy}}(t) - P_{\text{buy,dh}}(t))^2 + \varphi (W_n(t) - W_{n,\text{dh}}(t))^2 \right\} \quad (11)$$

In addition, in the model described in Eq. (1), to ensure that the scheduling scheme does not lose rationality while guaranteeing robustness and economy, the probability of each scene is constrained using a combined parametric constraint consisting of a 1-parametric condition and an ∞ -parametric condition [39]:

$$\Omega = \left\{ \{p_n\} \left| \begin{array}{l} p_n \geq 0, n = 1, 2, \dots, N \\ \sum_{n=1}^N p_n = 1 \\ \sum_{n=1}^N |p_n - p_{0,n}| \leq \theta_1 \\ \max_{1 \leq n \leq N} |p_n - p_{0,n}| \leq \theta_\infty \end{array} \right. \right\} \quad (8)$$

where $p_{0,n}$ is the initial probability distribution of the reduced historical scenes of the scenic load scenes; θ_1 is the allowable deviation of the probability of the actual scenes for the 1-paradigm condition, and θ_∞ is the allowable deviation of the probability of the actual scenes for the ∞ -paradigm condition.

From the literature [40] it can be shown that the 1-parametric and ∞ -parametric conditions in Eq. (8) satisfy the confidence set:

$$\left\{ \begin{array}{l} \Pr \left\{ \sum_{n=1}^N |p_n - p_{0,n}| \leq \theta_1 \right\} \geq 1 - 2Ne^{-\frac{2M\theta_1}{N}} \\ \Pr \left\{ \max_{1 \leq n \leq N} |p_n - p_{0,n}| \leq \theta_\infty \right\} \geq 1 - 2Ne^{-2M\theta_\infty} \end{array} \right\} \quad (9)$$

where M is the number of historical scenes.

Let the right-hand side of Eq. (9) be equal to α_1 and α_∞ , then we have:

$$\left\{ \begin{array}{l} \theta_1 = \frac{N}{2M} \ln \frac{2N}{1 - \alpha_1} \\ \theta_\infty = \frac{1}{2M} \ln \frac{2N}{1 - \alpha_\infty} \end{array} \right. \quad (10)$$

Thus, Eqs. (8) and (10) form the feasible domain for the probability distribution of the worst scenario for scenic loads.

3.2. Intra-day rolling optimal scheduling

The intra-day rolling optimization mainly takes 4 h as the control time domain and 15 min as the index value. After updating the intra-day source load forecast information under the above time domain, the intra-day rolling optimization scheduling plan is further developed according to the day-ahead scheduling plan, while providing reference for the real-time adjustment phase.

Due to the long response time scales of energy storage devices compared to other energy transmission devices, and frequent start-ups and shutdowns can significantly reduce the service life of energy storage devices. Therefore, in order to avoid the constant start-stop of energy storage equipment to ensure the service life of the equipment, and

In Eq. (11), T is the starting moment of each intra-day optimal scheduling; ΔT is the intra-day optimal scheduling time scale; k is the number of intra-day scheduling periods; $P_{\text{buy,dh}}(t)$ and $P_{\text{buy}}(t)$ denote the contact line power of the microgrid day-ahead plan and intra-day plan, respectively; $I_{n,\text{dh}}(t)$ and $I_n(t)$ are 0–1 variables, denoting the start-stop state of the n th energy storage device day-ahead plan and intra-day plan at time t ; C_n is the start-stop variation penalty cost of the n th energy storage device day-ahead plan and intra-day plan at time t , respectively; $W_{n,\text{dh}}(t)$ and $W_n(t)$ are the energy storage of the n th energy storage device at time t of the day-ahead and intra-day plans, respectively; σ and φ are the contact line power and energy storage variation penalty coefficients, respectively.

It should be noted that once the intra-day rolling optimization schedule has optimized the output of the equipment in the control time domain while meeting the constraints of the system, only the first 15-minute schedule is issued to complete the real-time adjustment stage; after completing this stage of intra-day rolling optimization, the optimization window is shifted forward by 15 min and the optimization is carried out again based on the latest source load forecast information. In this way, the optimization is repeated until the whole intra-day rolling optimization phase is completed.

3.3. Real-time adjustment optimal scheduling

Since the prediction accuracy of WT, PV and load all have the characteristic of improving as the time scale decreases, the ultra-short-term prediction values can be used to adjust the microgrid for optimal scheduling in real time.

As the last step of the entire microgrid multi-time scheduling optimization, the real-time adjustment stage takes 15 min as the control time domain and 5 min as the index value. The real-time adjustment stage and intra-day rolling optimization scheduling are nested and executed, with multiple real-time adjustment stages nested in each intra-day scheduling plan. When the intra-day rolling optimization of a certain time period is completed, multiple real-time adjustment optimizations need to be completed based on the issued intra-day rolling scheduling plan before the next stage of intra-day rolling optimization scheduling can be carried out. Repeat the above process until the completion of the final capacity adjustment plan of the system.

Different from the traditional centralized MPC, the DMPC algorithm based on the idea of multiple intelligences decomposes the system into smaller subsystems, and divides the overall task among the intelligences by exchanging information among the subsystems, which can ensure the control performance of the system while reducing the complexity of the problem. Therefore, this paper adopts the DMPC algorithm for real-time adjustment optimal scheduling plan.

According to the requirements of the DMPC algorithm, the prediction model of the system needs to be established, and the prediction model of the microgrid in this paper is shown in the state space equation described in Eq. (12).

$$\begin{bmatrix} P_{gt}(t+\Delta t|t) \\ P_{gb}(t+\Delta t|t) \\ P_{p2g}(t+\Delta t|t) \\ P_{buy}(t+\Delta t|t) \\ G_{buy}(t+\Delta t|t) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} P_{gt}(t) \\ P_{gb}(t) \\ P_{p2g}(t) \\ P_{buy}(t) \\ G_{buy}(t) \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \\ \frac{1}{\eta^{gt}} & \frac{1}{\eta^{gb}} & \frac{\eta^{p2g}}{H_{gas}} \end{bmatrix} \begin{bmatrix} \Delta P_{gt}(t) \\ \Delta P_{gb}(t) \\ \Delta P_{p2g}(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta P_{wt}(t) \\ \Delta P_{pv}(t) \\ \Delta L_e(t) \\ \Delta L_h(t) \\ \Delta L_g(t) \end{bmatrix} \quad (12)$$

In Eq. (12), $P_{gt}(t+\Delta t|t)$, $P_{gb}(t+\Delta t|t)$, $P_{p2g}(t+\Delta t|t)$, $P_{buy}(t+\Delta t|t)$ and $G_{buy}(t+\Delta t|t)$ are the final scheduling plan values for gas turbines, gas boilers, P2G, microgrid electricity purchases and gas purchases from the gas network after real-time adjustment; $P_{gt}(t)$, $P_{gb}(t)$, $P_{p2g}(t)$, $P_{buy}(t)$ and $G_{buy}(t)$ are the scheduling plan values for gas turbines, gas boilers, P2G, microgrid electricity purchases and gas purchases from the gas network in the previous stage; $\Delta P_{gt}(t)$, $\Delta P_{gb}(t)$ and $\Delta P_{p2g}(t)$ are the revised values for gas turbines, gas boilers and electricity to gas respectively in the real-time phase; $\Delta P_{wt}(t)$ and $\Delta P_{pv}(t)$ are the WT and PV power output disturbances; $\Delta L_e(t)$, $\Delta L_h(t)$ and $\Delta L_g(t)$ are the perturbations of electrical, thermal and gas loads.

Subsequently, based on the prediction model developed in Eq. (12), the objective function which aims at minimizing the difference between the output vector and the target vector, as well as the variables of the control vector, is set up as follow [41]:

$$\min J_i = (Y_{i,fore} - Y_{i,ref})^T W_i (Y_{i,fore} - Y_{i,ref}) + U_i^T Q_i U_i \quad (13)$$

$$Y_{i,fore} = \begin{bmatrix} P_{buy}^i(t+\Delta t|t) & \cdots & P_{buy}^i(t+m\Delta t|t) \\ G_{buy}^i(t+\Delta t|t) & \cdots & G_{buy}^i(t+m\Delta t|t) \end{bmatrix} \quad (14)$$

$$Y_{i,ref} = \begin{bmatrix} P_{buy}^{i,ref}(t+\Delta t|t) & \cdots & P_{buy}^{i,ref}(t+m\Delta t|t) \\ G_{buy}^{i,ref}(t+\Delta t|t) & \cdots & G_{buy}^{i,ref}(t+m\Delta t|t) \end{bmatrix} \quad (15)$$

$$U_i = \begin{bmatrix} \Delta P_{gt}^i(t+\Delta t|t) & \cdots & \Delta P_{gt}^i(t+m\Delta t|t) \\ \Delta P_{p2g}^i(t+\Delta t|t) & \cdots & \Delta P_{p2g}^i(t+m\Delta t|t) \\ \Delta P_{gb}^i(t+\Delta t|t) & \cdots & \Delta P_{gb}^i(t+m\Delta t|t) \end{bmatrix} \quad (16)$$

In Eqs. (13)–(16), $Y_{i,fore}$ is the predicted output vector composed of the purchased electricity and purchased gas values of the i th subsystem; $Y_{i,ref}$ is the reference output vector composed of the purchased electricity and purchased gas reference values of the i th subsystem; U_i is the adjustment quantity of the i th subsystem; W_i and Q_i are the weight matrices of the output and control quantities respectively; m is the prediction step.

The steps of the algorithm in solving the objective function are shown in detail in Fig. 4:

In addition, the real-time adjustment phase needs to satisfy not only the prediction model described in Eq. (12), but also the upper and lower adjustment constraints for the coupled devices, as shown in Eqs. (17)–(22).

$$\Delta P_{gt}^{\min} \leq \Delta P_{gt}(t+k\Delta t|t) \leq \Delta P_{gt}^{\max} \quad (17)$$

$$P_{gt}^{\min} \leq P_{gt}(t+k\Delta t|t) \leq P_{gt}^{\max} \quad (18)$$

$$\Delta P_{p2g}^{\min} \leq \Delta P_{p2g}(t+k\Delta t|t) \leq \Delta P_{p2g}^{\max} \quad (19)$$

$$P_{p2g}^{\min} \leq P_{p2g}(t+k\Delta t|t) \leq P_{p2g}^{\max} \quad (20)$$

$$\Delta P_{gb}^{\min} \leq \Delta P_{gb}(t+k\Delta t|t) \leq \Delta P_{gb}^{\max} \quad (21)$$

$$P_{gb}^{\min} \leq P_{gb}(t+k\Delta t|t) \leq P_{gb}^{\max} \quad (22)$$

3.4. Process of multi-time scale optimal scheduling model

The microgrid multi-time scale scheduling method can be divided into day-ahead stage, intra-day stage and real-time stage. The detailed process is shown in Fig. 5.

4. Case study

4.1. Input data and simulation setup

The parameters of the microgrid studied in this paper are shown in Table 1 [42]. The predicted value of WT, PV and load in the case are shown in Fig. 6, and the electric price is shown in Fig. 7. The natural gas price is 1.8¥/m³, and the low thermal value is 9.7 kWh/m³. The CPLEX commercial solver is used under MATLAB 2016a for modeling and solving. In the day-ahead stage, the confidence level of both the 1-parametric constraint(α_1) and the ∞ -parametric constraint(α_∞) in Eq. (10) is set to 0.9, and 10 sets of initial scenarios are obtained from 5000 sets of historical data by forward back substitution cuts. At the same time, this paper sets the predicted data of the intra-day rolling optimal scheduling stage and the real-time adjusted optimal scheduling stage to be generated from the day-ahead prediction data and the error

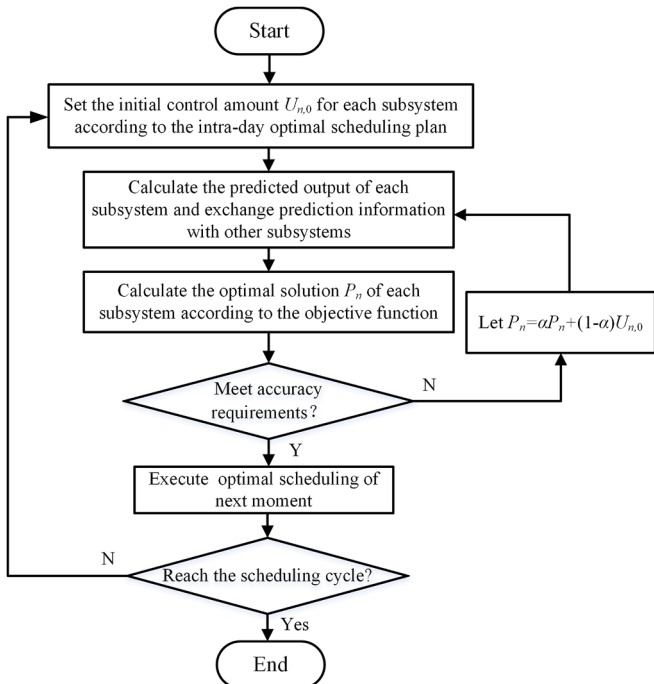


Fig. 4. The flowchart of DMPC calculation.

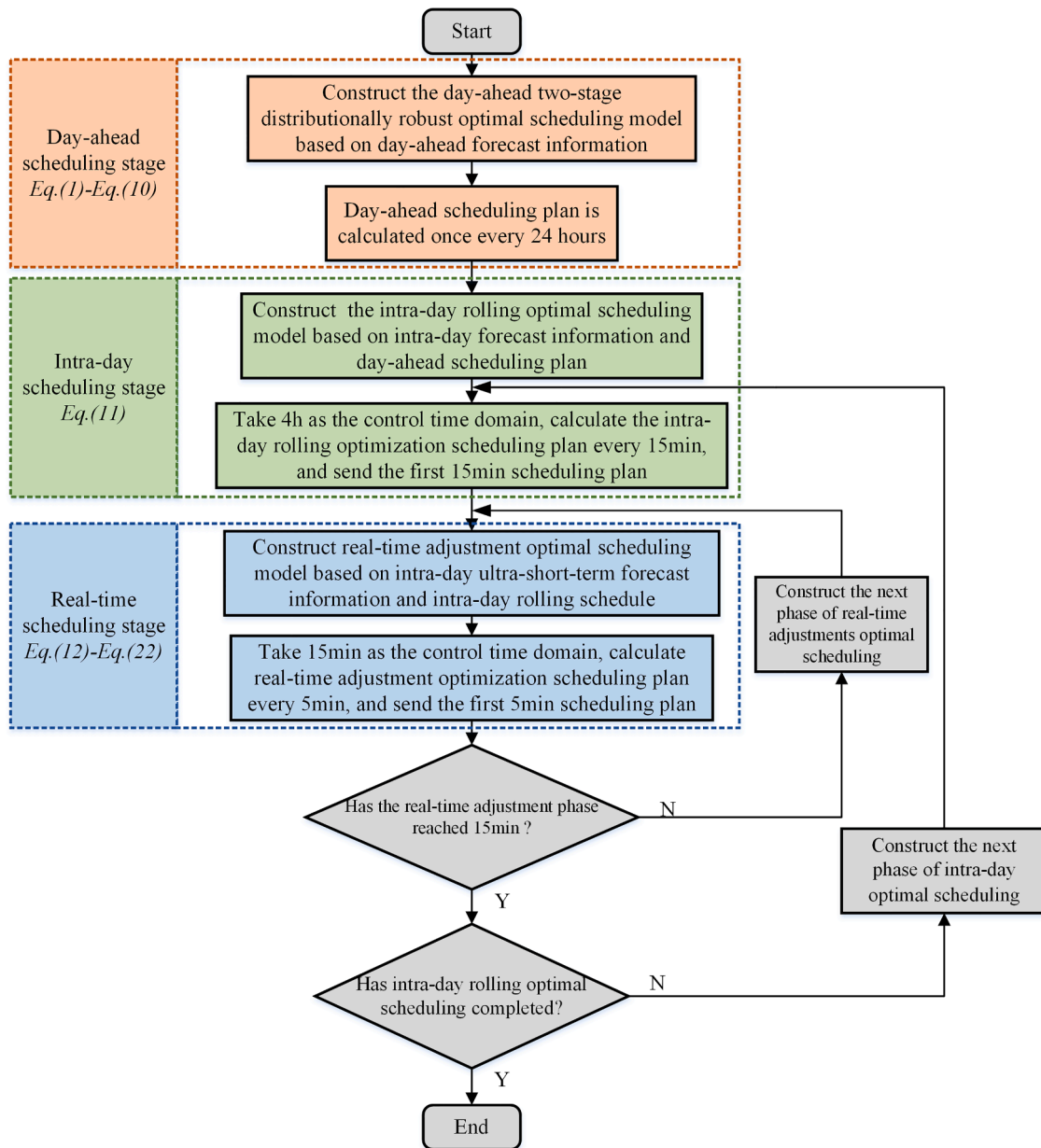


Fig. 5. The flowchart of multi-time scale optimal scheduling.

distributions they obey. The error distributions of WT, PV and loads are assumed to obey Beta distribution, Weibull distribution and normal distribution, respectively, and the uncertainty levels of each part are shown in Table 2 [43].

4.2. Analysis of day-ahead two-stage distributionally robust optimal scheduling

On the basis of the day ahead forecast value of WT, PV and load, according to the day ahead two-stage distributionally robust optimization objective function and the constraints of the model established in Section 3.1, the day ahead scheduling plan of the microgrid is calculated by the scale value of 1 h. The results are shown in Figs. 8–10.

(1) Day-ahead optimal scheduling of electrical energy

The results of the day-ahead optimized scheduling of electricity are shown in Fig. 8.

The battery is charged during the lower price hours of 3–5 h and discharged during the higher price hours of 11 h, 14 h and 17 h, so as to reduce the cost of electricity and fill the valleys by taking advantage of the peak-to-valley characteristics of electricity prices.

The working time of the P2G equipment is mainly concentrated in the 1–8 h and 19–24 h, because the wind power is abundant and the demand for electricity is not high during these two hours, the P2G equipment can achieve further consumption of wind power, thus reducing the cost of gas purchase.

The gas turbine starts working in the 7 h and produce more power in the 12–20 h when the electricity price is high, so as to avoid the microgrid buying power in this period, which makes the system's energy purchase cost higher, and increases the power sales to the grid of microgrid.

As the electricity price is low in the 2–9 h and the electricity transmission devices and batteries have electricity demand, the microgrid's power purchase is mainly concentrated in this period. At the same time, the microgrid is able to sell excess electricity during the high tariff

Table 1
Parameters of microgrid.

Name	Parameters	values	Name	Parameters	values
Gas Turbine	Electrical efficiency	0.35	Battery	Charging efficiency	0.9
	Heat efficiency	0.45		Discharging efficiency	0.9
	Max capacity	2000kW		Max capacity	1600kWh
	Min capacity	0kW		Min capacity	100kWh
	Max climbing power	800kW		(Dis)Charging power capacity	500kW
	μ	0.15		ν	0.05
Gas Boiler	π	0.12	Heat Storage Tank	start-stop variation penalty cost	0.04
	Heat efficiency	0.8		Charging efficiency	0.95
	Max capacity	1300kW		Discharging efficiency	0.95
	Min capacity	0kW		Max capacity	1200kWh
	Max climbing power	600kW		Min capacity	100kWh
	μ	0.12		(Dis)Charging power capacity	450kW
P2G	π	0.1	Gas Storage Tank	ν	0.02
	Gas efficiency	0.6		start-stop variation penalty cost	0.01
	Max capacity	300kW		Charging efficiency	0.98
	Min capacity	0kW		Discharging efficiency	0.98
	Max climbing power	300kW		Max capacity	1000m ³
	μ	0.1		Min capacity	50m ³
	π	0.08		(Dis)Charging power capacity	200m ³
				ν	0.03
				start-stop variation penalty cost	0.02

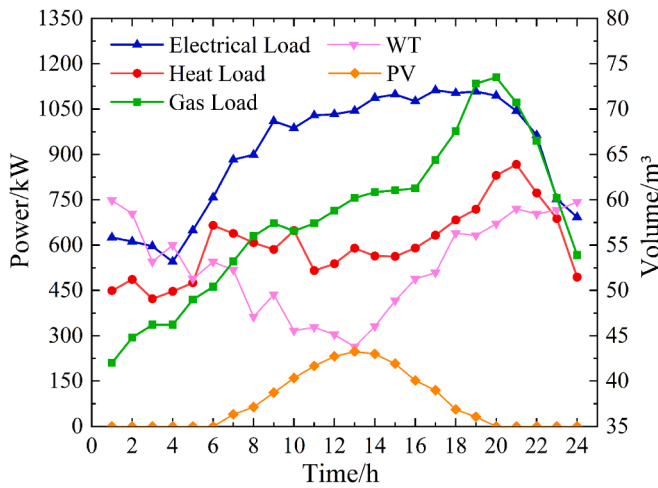


Fig. 6. The predicted value of WT, PV and Load.

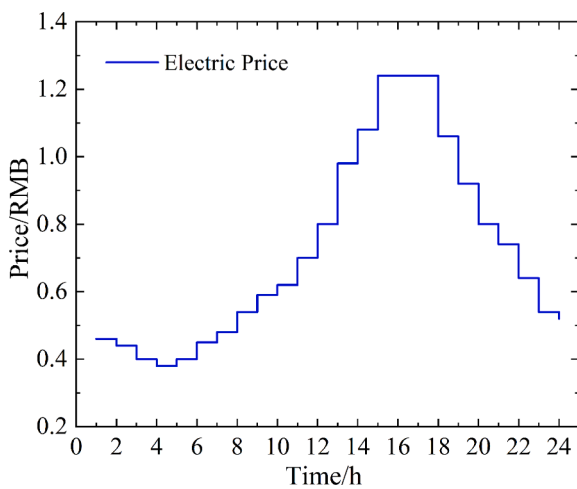


Fig. 7. Electric price.

Table 2
Level of uncertainty of predicted values in different scheduling stage.

Scheduling Stage	Level of uncertainty				
	PV	WT	Electrical Load	Heat Load	Gas Load
Intra-day Scheduling	0.2	0.25	0.2	0.1	0.15
Real-time Scheduling	0.15	0.2	0.1	0.05	0.12

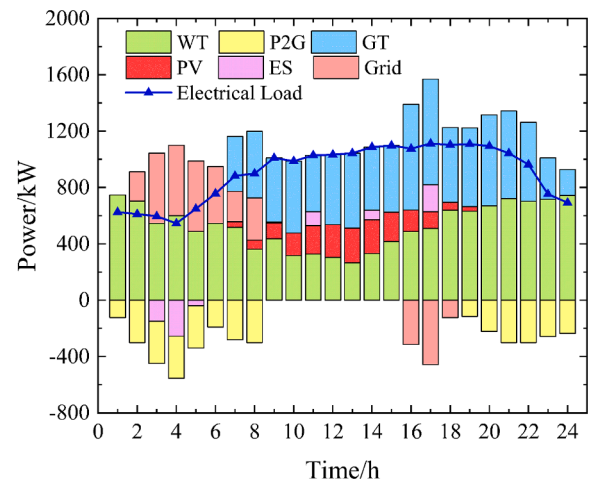


Fig. 8. Day-ahead electrical energy optimal scheduling result.

period 16–18 h to generate revenue, while meeting the demand of the electrical load.

(2) Day-ahead optimal scheduling of thermal energy

The results of the day-ahead optimized scheduling of heat are shown in Fig. 9.

The heat storage tank stores heat during the low heat demand hours (10–17 h), enabling it to release heat during 1 h, 6–7 h and 21–23 h to reduce the amount of purchased gas required for heat production by the gas turbine and gas boiler.

Since the gas turbine will generate a large amount of waste heat when supplying heat during 7–20 h, the heat load during these hours is mainly met by the gas turbine, while the heat load of the remaining

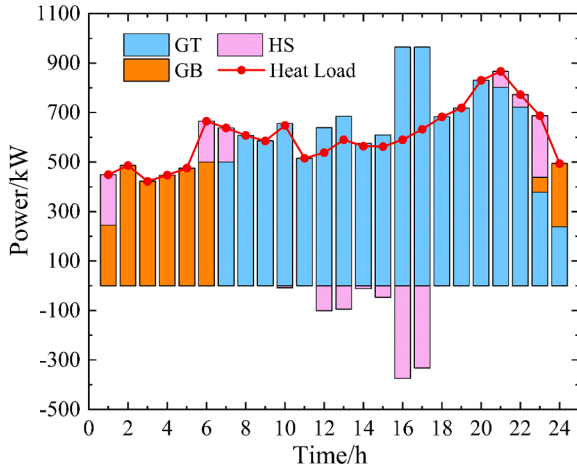


Fig. 9. Day-ahead heat energy optimal scheduling result.

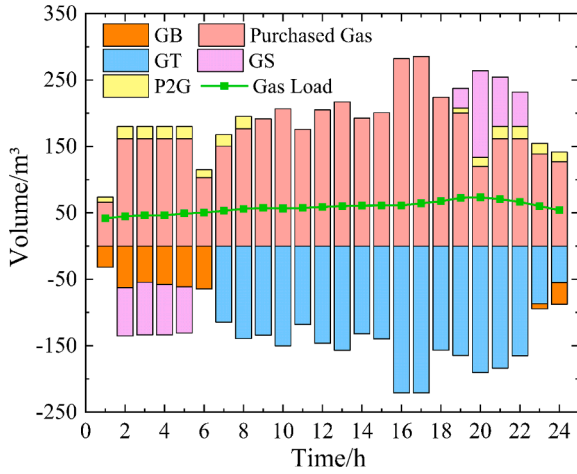


Fig. 10. Day-ahead gas energy optimal scheduling result.

hours when the gas turbine is short of heat is mainly met by the gas boiler.

(3) Day-ahead optimal scheduling of natural gas

The results of the day-ahead optimized scheduling of natural gas are shown in Fig. 10.

The microgrid mainly uses gas storage tanks to store gas during the 2–5 h when WT, PV output is high and electricity prices are low.

At the same time, P2G is used to convert electricity to natural gas during the 1–8 h and 19–24 h, which not only provides space for wind power consumption, but also further reduces the system's energy purchase cost.

Meanwhile, the microgrid purchases gas and uses gas storage tanks to release gas during the 19–22 h when electricity and heat load demand is high to meet the need of gas boiler, gas turbines and gas loads.

Table 3

Comparison of different day-ahead optimal scheduling strategy.

Day-ahead optimal scheduling strategy	Cost of day-ahead scheduling /¥	Solving time/s
Day-ahead deterministic optimal scheduling	15,386.08	5.12
Day-ahead two-stage distributionally robust optimal scheduling	15,864.72	32.46

(4) Comparison of previous day's optimal scheduling options

The economics and solution time of the proposed day-ahead two-stage distributionally robust optimal scheduling and day-ahead deterministic optimal scheduling are compared in Table 3:

As can be seen from Table 3, the day-ahead scheduling cost of the proposed day-ahead two-stage distributionally robust optimization is ¥478.64 higher than the day-ahead deterministic optimization. The reason is that, unlike deterministic optimization, the two-stage distributionally robust optimization method takes into account the probability of worst-case WT, PV and load distribution, which increases the cost by 3.11% compared to deterministic optimization. This makes the day-ahead scheduling plan more robust, thus allowing the microgrid to be scheduled intra-day without incurring excessive costs due to extreme scenarios. In addition, the solution time is longer due to the fact that the day-ahead two-stage distributionally robust optimization scheduling takes into account multiple scenarios and scenario probability distribution values, but still meets the requirements of the day-ahead scheduling plan.

4.3. Analysis of intraday energy storage equipment optimal scheduling

In order to avoid the frequent start-up and shutdown of the energy storage devices, the energy storage devices only participate in the day-ahead two-stage distributionally robust optimization scheduling and intra-day rolling optimization scheduling. The power variation of the energy storage equipment in the intra-day stage is mainly calculated by Eq. (11) and the intra-day source load forecast information.

Figs. 11–13 show the power variation curves of three different types of energy storage devices before and during the day. The fluctuation of scenic load in the real-time adjustment phase is mainly regulated and smoothed by gas turbine, gas boiler and P2G.

As can be seen in Figs. 11–13, the start-stop status and power variation trends of each energy storage device in the intra-day rolling optimal scheduling phase are generally similar to those in the previous two-stage distributionally robust optimal scheduling. This is mainly because most of the fluctuations caused by renewable energy and load can be smoothed out by each energy transmission device in the real-time adjustment phase, and the remaining part can be adjusted by the energy storage devices to reduce the overall operating costs.

4.4. Analysis of energy transmission equipment and contact line scheduling based on the DMPC algorithm

In the real-time adjustment stage, based on the obtained day-ahead

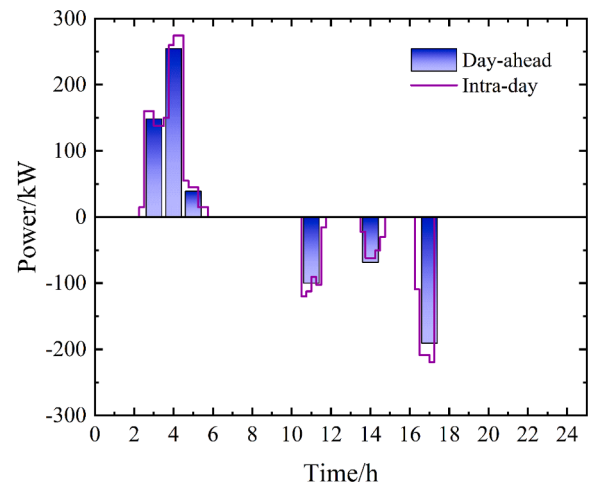


Fig. 11. Day-ahead and Intra-day power variation of battery.

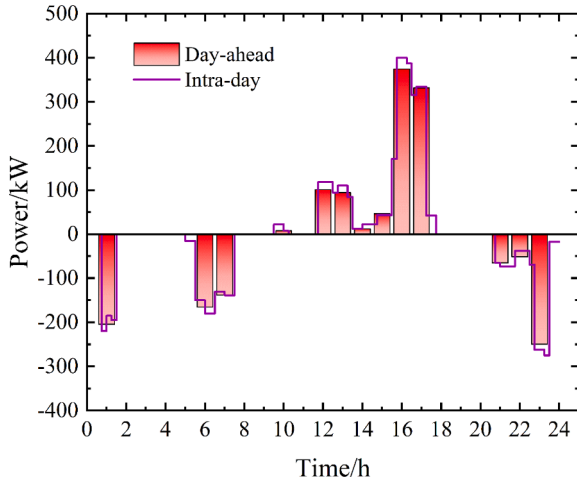


Fig. 12. Day-ahead and Intra-day power variation of heat storage tank.

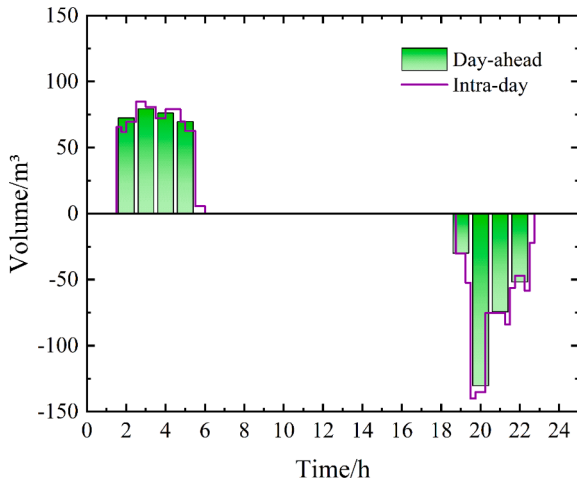


Fig. 13. Day-ahead and Intra-day power variation of gas storage tank.

and intra-day scheduling scheme, the adjustment power of gas turbine, gas boiler and P2G equipment is calculated according to the prediction model described in Eq. (12) and the optimization model described in Eqs. (13)–(22).

(1) Analysis of the results of real-time phase scheduling of energy transmission equipment

In order to verify the effectiveness and feasibility of the DMPC algorithm in this paper during the real-time adjustment phase, the output curves of the gas turbine, P2G and gas boiler under the real-time adjustment phase are shown in Figs. 14–16.

As can be seen in Figs. 14–16, both the MPC and DMPC control methods are able to achieve real-time adjustment of each energy transmission device based on certain prediction models, and the change trend of each energy transmission device is basically the same under both control methods. However, the power fluctuation of each energy transmission device under the centralized MPC control method is relatively larger and there are several obvious "spikes" in the real-time adjustment plan. This is mainly due to the fact that the centralized MPC control method sets the objective function to solve the whole system as a whole to reach the overall optimal objective and the model is solved with high order and large dimensionality, which cannot take into account the control performance and adjusted power of each energy-coupled device. The DMPC method rewrites the objective function of

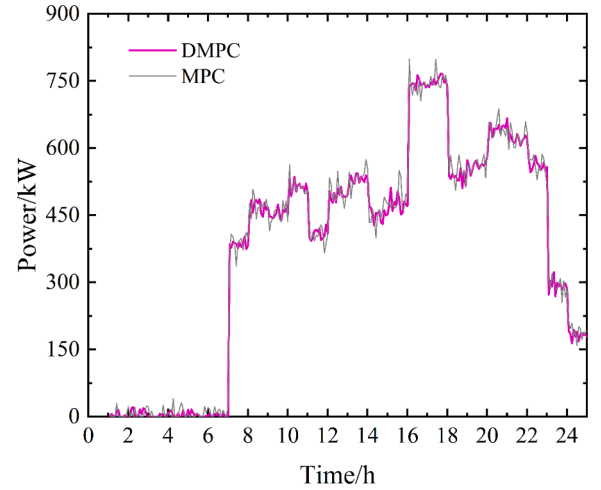


Fig. 14. Real-time power variation of GT.

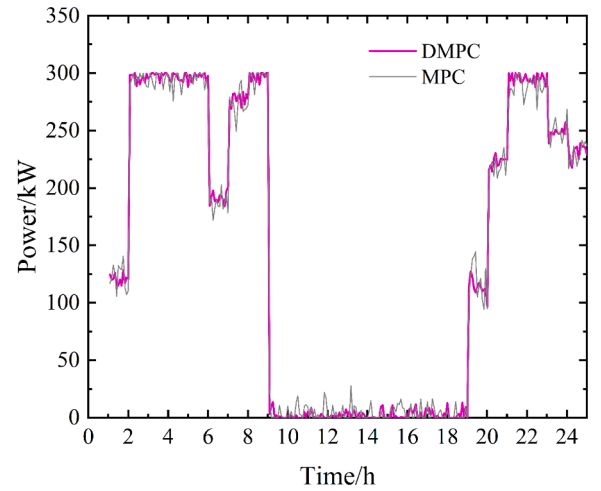


Fig. 15. Real-time power variation of P2G.

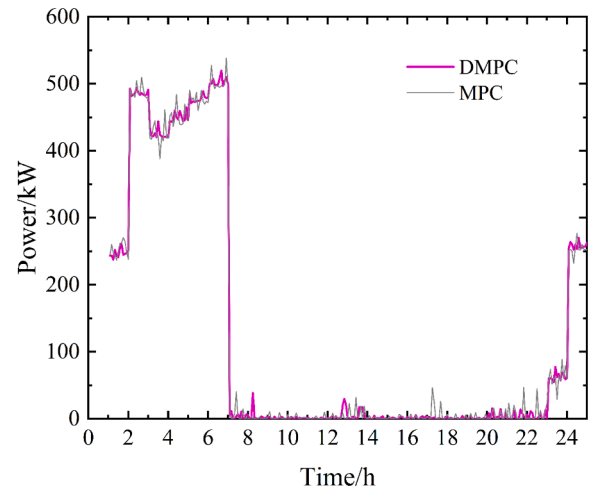


Fig. 16. Real-time power variation of GB.

the centralized MPC method and decomposes the optimal scheduling task of the system into individual sub-tasks and distributes them to each sub-system. Each subsystem not only needs to achieve its own tracking

and optimal control of the day-ahead and intra-day schedules, but also needs to achieve iterative calculation through information exchange with other subsystems, thus ensuring the optimal control performance of the entire microgrid system. Compared with the centralized MPC algorithm, the power output of energy transmission equipment under DMPC is more stable and less prone to change, which avoids the loss caused by frequent start/stop and power changes of the equipment to a certain extent and effectively improves the economic efficiency and energy supply stability of the microgrid.

(2) Analysis of contact line scheduling results

To further demonstrate the advantages of the DMPC algorithm in ensuring the stability of the microgrid's contact line power with the external grid and gas network, this paper compares and analyses the microgrid's contact line power with the external network under DMPC, MPC and the Day-ahead scheduling (DAS) control algorithm, as shown in Figs. 17–20. Where DAS means that instead of using intra-day rolling scheduling optimization and real-time adjustment scheduling optimization, the microgrid directly smooths out the errors caused by the day-ahead forecast through power and gas purchases on the basis of the contact line power in the day-ahead scheduling.

Both the DMPC control method and the MPC control method can not only reduce the power fluctuations effectively between the microgrid and the external grid and the external gas network, but also achieve effective tracking of the scheduling plan of the previous contact line when forecast errors cannot be avoided, ensuring the safety and reliability of the contact line.

In order to quantitatively analyze the tracking effect, the deviation ratio of contact line is used for specific characterization, which is defined as below:

$$R = \frac{\sqrt{\sum_{t=1}^T (L_{rt,t} - L_{da,t})^2}}{\sqrt{\sum_{t=1}^T L_{da,t}^2}} \quad (23)$$

where, $L_{rt,t}$ and $L_{da,t}$ are the value of contact line in real-time scheduling stage and day-ahead scheduling stage.

According to Eq. (23), tracking deviation ratios of contact line of grid and gas network under different strategies are shown in Table 4:

As is shown in Table 4, the tracking deviation ratios in the contact line under the DMPC control method are 1.76% and 1.28%, which is 3.23% and 2.45% lower than those under the MPC control method. This is mainly because under the DMPC control method, each energy

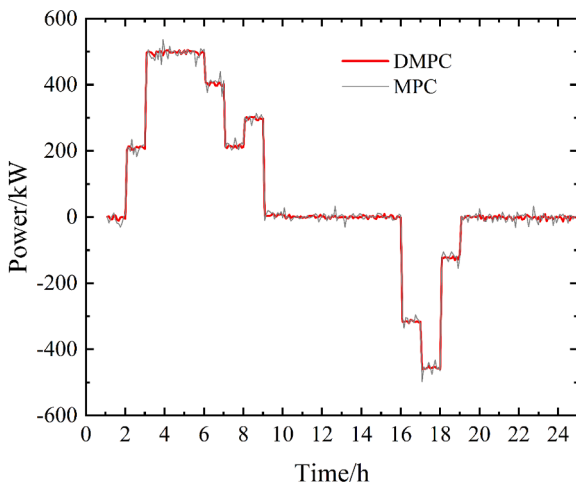


Fig. 17. Variation of purchased power and sell power of DMPC and MPC.

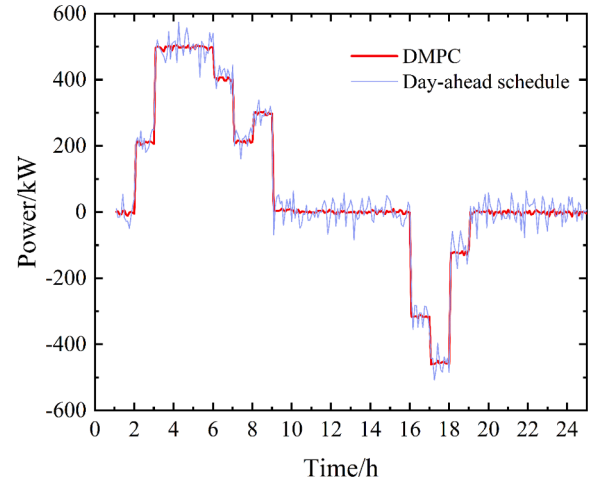


Fig. 18. Variation of purchased power and sell power of DMPC and DAS.

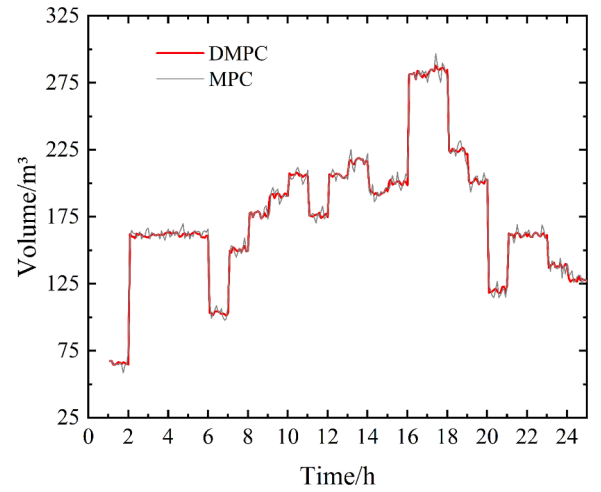


Fig. 19. Variation of purchased gas of DMPC and MPC.

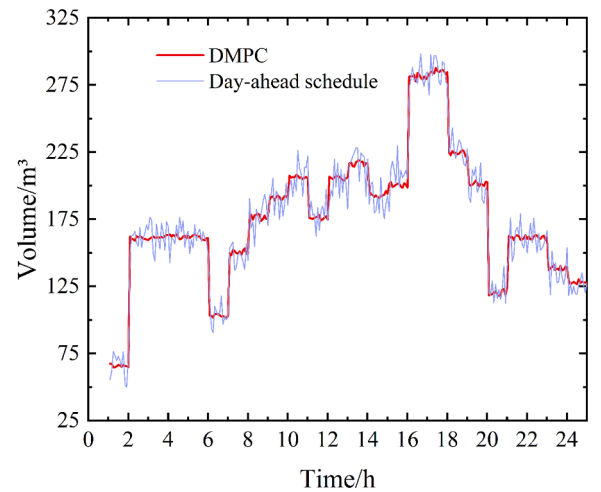


Fig. 20. Variation of purchased gas of DMPC and DAS.

transmission device coordinates with each other and is able to respond more quickly to the fluctuations caused by the uncertainty of the WT, PV and load, making it easier to achieve its own optimum and the system

Table 4
Comparison of tracking deviation ratio of contact line under different strategies.

	DMPC	MPC	DAS
Grid	1.76%	4.99%	10.53%
Gas network	1.28%	3.73%	7.20%

optimum. This can strengthen the internal and overall coordination of the system and effectively suppressing the fluctuations in the contact line. Although the MPC control method can achieve contact line power tracking in microgrids, the effect is more volatile than that of DMPC; this is mainly due to the fact that MPC can only guarantee the overall optimum by solving centrally and the system coordination is not strong enough to guarantee the optimal performance of each coupling device. On the other hand, the deviation ratio of contact line under DAS strategy is the highest, up to 10.53% and 7.20%, the reason is that DAS only relies on the day-ahead scheduling plan and does not rely on individual energy transmission equipment for regulation. This is not conducive to the stable operation of the contact line, and also has the possibility of reducing the service life of the equipment to a certain extent, which may influence the energy supply of the microgrid under a long time.

(3) Intraday scheduling cost and total cost analysis

The intraday scheduling cost and total cost of the three methods are shown in Table 5. The intra-day scheduling cost and the total cost of the microgrid under the MPC control method are reduced compared to the DAS strategy, which indicates that the MPC control method is able to reduce the cost by scheduling energy transmission devices to smooth out the forecasting errors.

However, compared with the DMPC control method, the overall coordination capability of the system under the MPC control method is lower, and the control effect is slightly inferior to that of DMPC, thus the cost is higher than that of DMPC. The intra-day scheduling and total cost of the microgrid under the DMPC control method are lower than those of the other two methods, mainly due to the coordination among the subsystems under this method. Therefore, the final operating cost is the lowest and the performance in contact line power tracking is better, which can effectively avoid the impact of source-load uncertainty on the optimal scheduling of the microgrid.

4.5. Analysis of overall multi-time scale scheduling results

To fully demonstrate the effectiveness of the overall multi-time scale optimal scheduling strategy proposed in this paper, two other scheduling strategies are used in this section and are compared and analyzed with three datasets with different uncertainty levels. The three data sets correspond to different uncertainty levels of low, medium and high, and each data set contains 500 different specific scenarios of intra-day source load forecasts. Each specific scenario in the dataset is randomly generated by the predicted value according to the fluctuation thresholds percentage, which is subject to the probability distribution in Section 4.1. The fluctuation threshold percentages for WT, PV and load at different uncertainty levels are shown in Table 6 [44].

Based on the data sets with different levels of uncertainty to which Table 6 belongs, the following three different scheduling strategies were compared and analyzed.

M1: Single day-ahead two-stage distributionally robust optimal

Table 5
Comparison of different real-time scheduling strategy.

Real-time scheduling strategy	Cost of intra-day scheduling /¥	Total cost/¥
Day-ahead schedule	1447.26	17,311.98
MPC	928.51	16,793.23
DMPC	712.04	16,576.76

Table 6
Fluctuation thresholds percentage under different uncertainty levels.

Scheduling Stage	Level of uncertainty	Fluctuation thresholds percentage/%				
		PV	WT	Electrical Load	Heat Load	Gas Load
Intra-day Scheduling	Low	10	15	10	5	10
	Medium	20	25	20	10	15
	High	30	40	40	20	30
Real-time Scheduling	Low	5	10	5	5	6
	Medium	10	15	10	10	12
	High	25	20	15	15	24

scheduling without intra-day scheduling

Day-ahead scheduling plan is carried out using two-stage distributionally robust optimization to deal with the uncertainty of distributed energy and load. But intra-day rolling optimal scheduling and real-time adjustment optimal scheduling are not constructed.

M2: Day-ahead deterministic optimized scheduling, intra-day rolling optimal scheduling and real-time adjustment optimal scheduling based on DMPC

The scheduling plan is based on the source load forecasts on three different time scales: day-ahead, intra-day and real-time. The scheduling plan is developed using DMPC for intra-day and real-time adjustments, but the day-ahead does not take into account the uncertainty of scenery loads and only uses deterministic optimization strategies.

M3: The proposed day-ahead two-stage distributionally robust optimal scheduling, intra-day rolling optimal scheduling and real-time adjustment optimal scheduling based on DMPC

The average cost and cost growth rates of three different scheduling approaches mentioned above are calculated based on each specific scenario in the generated dataset of different uncertainty levels. The average costs and their cost growth rates for the above three scheduling strategies for different uncertainty level data sets are shown in Table 7.

As can be seen from Table 7, M1 strategy has the highest average cost at different levels of uncertainty and the fastest cost growth among the three strategies when the level of uncertainty is gradually increased. This is mainly due to the fact that although the single day-ahead two-stage distributionally robust optimal scheduling only takes into account certain source-load uncertainty in the day-ahead stage, it lacks the support of intra-day source-load forecast values with higher accuracy and further optimization for intra-day rolling and real-time adjustment. This makes the strategy unable to fully counter the impact of unplanned instantaneous fluctuations in source-load power on the energy management of multi-energy complementary microgrids, resulting in high final scheduling costs in certain extreme scenarios. The final scheduling cost is high and the overall robustness is poor.

In comparison, M2 strategy has a lower average cost and cost growth rate compared to the M1 strategy. Thanks to the forecast data of source and load in the day-ahead and real-time phases, the microgrid under the M2 strategy can respond to unplanned instantaneous power fluctuations of source and load that are not taken into account in the day-ahead phase. However, as source load uncertainty is not taken into account in the day-ahead phase, excessive system adjustments may occur in the intra-day and real-time phases due to adjustment needs and supply-demand balance in extreme scenarios, thus increasing the scheduling cost of the microgrid.

In contrast, M3 strategy is the lowest in terms of cost growth rate

Table 7
Average cost and cost growth rate under different scheduling strategies.

Method	Average cost /¥			Cost growth rate /%		
	Low	Medium	High	Low	Medium	High
M1	16,456.32	17,306.14	18,796.25	\	5.16	8.61
M2	16,194.07	16,885.72	18,120.93	\	4.27	7.32
M3	16,013.42	16,572.37	17,643.58	\	3.49	6.46

compared to the other two strategies, while achieving the optimal average operating cost. Although the day-ahead cost of this strategy is higher than deterministic optimization as described in Table 4, the consideration of day-ahead source load uncertainty makes the day-ahead schedule robust. The intraday and real-time scheduling plans based on the day-ahead plan therefore have a lower average adjustment cost in the dataset under different levels of uncertainty, improving the microgrid's ability to cope with unplanned instantaneous power fluctuations. This makes the overall scheduling strategy less costly than the M2 strategy. In other words, the multi-time scale scheduling strategy proposed in this paper can reduce the cost of microgrid scheduling while effectively ensuring that the microgrid has strong robustness in operation, which in turn can improve the economy, security and stability of the microgrid.

5. Conclusion

In order to effectively cope with the uncertainty problem of source and load in microgrids, this paper proposes a multi-time scale optimal scheduling strategy for microgrids considering source and load uncertainty, and analyses the proposed multi-time scale scheduling strategy by using specific arithmetic examples. The conclusions can be summarized as follows:

- (1) The day-ahead two-stage distributionally robust optimal scheduling model provides a robust day-ahead schedule by developing a day-ahead schedule for the worst scenery load probability distribution. Although the cost is 3.11% higher than that of the day-ahead deterministic optimal scheduling method, the improved robustness of the day-ahead scheduling plan can effectively reduce the adjustment costs in the intra-day and real-time phases.
- (2) The intra-day and real-time phases utilize the DMPC method to make adjustments based on the tracking day-ahead plan using energy storage devices and energy transmission devices to smooth out the fluctuations of source and load, which demonstrate good tracking capability in terms of contact line interaction power. Also compared with the DAS and MPC methods, the deviation ratio of contact line under the proposed method is the smallest, which is 1.76% and 1.28%. This shows that the proposed method can not only improve the economy of the intra-day and real-time scheduling scheme of the microgrid, but also enhance the friendliness of microgrid to external grids and gas networks.
- (3) Compared with the other two strategies, the proposed multi-time scale scheduling strategy considering uncertainty of source and load has the lowest cost growth rate, respectively 3.49%, and 6.46% under various uncertainty level scenarios. It further demonstrates that the proposed strategy can ensure good robustness on the basis of economic efficiency of the microgrid and effectively maintain the smooth power of contact lines, thus avoiding the impact of source-load uncertainty on the optimal scheduling of the microgrid.

In the future, the presented model will consider the stochastic feature of Demand Response Load in the microgrids and analyze its impact on the scheduling results. In addition, the energy flow between multiple microgrids in different areas can be considered to optimize the scheduling plan of multi-regional microgrids.

CRedit authorship contribution statement

Jianmin Hou: Conceptualization, Supervision, Project administration, Funding acquisition. **Wei jie Yu:** Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Visualization. **Zhihao Xu:** Formal analysis, Investigation. **Quanbo Ge:** Formal analysis, Investigation. **Zhi Li:** Resources, Data curation. **Ying Meng:**

Resources, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors do not have permission to share data.

Acknowledgments

This work is supported by the National Natural Science Foundation of China(71503136), and the Postgraduate Research & Practice Innovation Program of Jiangsu Province(SJCX22_0357).

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